

CHAPTER 10

10.1) ENTROPY, ENERGY, AND ENTHALPY OF VAN DER WAALS GAS

(a) By inserting (38) into (3.49):

$$\begin{aligned}\sigma &= -(\partial F / \partial \tau)_V \\ &= + N \{ \log [n_Q (V - Nb) / N] + 1 \} + N \tau (\partial \log n_Q / \partial \tau)_V \quad . \quad (S1)\end{aligned}$$

But $n_Q \propto \tau^{3/2}$ -- see (3.63) -- hence $\tau (\partial \log n_Q / \partial \tau) = 3/2$. If this is inserted into (S1), (73) results.

(b) $U = F + \tau \sigma$. Multiplying (73) by τ and adding the result to (38) cancels the logarithm and leads to

$$U = -N\tau + \frac{5}{2}N\tau - N^2 a / V = \frac{3}{2}N\tau - N^2 a / V \quad . \quad (74)$$

(c) From (39),

$$pV = \frac{N\tau}{1 - Nb/V} - \frac{N^2 a}{V} \cong N\tau(1 + Nb/V) - N^2 a / V \quad , \quad (S2)$$

where we have expanded the first term after the first equality into a geometric series, dropping all terms of higher order in b . From (74) and (S2):

$$H = U + pV = \frac{5}{2}N\tau + N^2 b \tau / V - 2N^2 a / V \quad . \quad (75)$$

This is a function of N , τ and V . To obtain H as a function of its natural independent variables N , τ and p , we must express V as a function of N , τ and p . Because all terms in (75) containing V are already correction terms to the first order in the van der Waals coefficients a and b , we obtain $H(\tau, p)$ to the first order by simply inserting the ideal gas law, $N/V = p/\tau$. If this is done, (75) goes over into (76). To solve the van der Waals equation of

state (40) for V and to insert the result into (75) would be pointless: The result would differ from (76) only by higher-order terms in a and b , which were neglected already in (75). Besides, the van der Waals approach is not valid to such a higher order in the first place.

10.2) CALCULATION OF dT/dp FOR WATER

We neglect the volume per molecule of the liquid relative to that of the vapor, and treat the vapor itself as an ideal gas -- Eqs. (16) and (17) -- in which case we may use the Clausius-Clapyron equation in the simplified form (20), which we write

$$\frac{dp}{dT} = \frac{Lp}{k_B T^2} = \frac{N_A L}{RT^2} p \quad . \quad (S1)$$

Here N_A is Avogadro's number and $R = N_A k_B = 8.31 \text{ J mol}^{-1} \text{K}$ is the gas constant.

In our problem we have $N_A L = 2260 \text{ J g}^{-1} \times 18 \text{ g mol}^{-1} = 4.07 \times 10^4 \text{ J mol}^{-1}$. With $T = 373 \text{ K}$ and $p = 1 \text{ atm}$: $dT/dp = 28.4 \text{ K/atm}$.

10.3) HEAT OF VAPORIZATION OF ICE

We make the same simplifications as in the solution to the preceding problem, to obtain again (S1) of the solution of the preceding problem. Solved for $N_A L$:

$$N_A L = \frac{RT^2}{p} \frac{dp}{dT} = -R \frac{d \log p}{d(1/T)} \quad . \quad (S1)$$

We have $d(\log p) = \log(4.58/3.88) = 0.1659$, $d(1/T) = 1/273 - 1/271 = -2.703 \times 10^{-5}$. Inserted into (S1), this leads to $N_A L = 5.10 \times 10^4 \text{ J mol}^{-1}$.

Comment: We have written the derivative in the form (S1) because in this form an accurate result for $N_A L$ is assured

even if the pressure difference Δp is not small compared to p itself: If N_{AL} were truly temperature-independent, the replacement of the differential in (S1) by finite differences would be exact.

10.4) GAS-SOLID EQUILIBRIUM

In three dimensions the energy states of a single oscillator are $(n_x + n_y + n_z)\hbar\omega - \varepsilon_0$, where n_x , n_y and n_z are independent non-negative integers and $-\varepsilon_0$ is again the ground state energy. The partition function for a single such oscillator is

$$\begin{aligned} Z_S &= \sum_{n_x} \sum_{n_y} \sum_{n_z} \exp\{-(n_x + n_y + n_z)\hbar\omega - \varepsilon_0\}/\tau\} \\ &= \exp(\varepsilon_0/\tau) \left\{ \sum_n [\exp(-n\hbar\omega/\tau)] \right\}^3 \\ &= \exp(\varepsilon_0/\tau) / [1 - \exp(-\hbar\omega/\tau)]^3, \end{aligned}$$

instead of (26). Thus, instead of (29):

$$\lambda_S \cong 1/Z_S = \exp(-\varepsilon_0/\tau) [1 - \exp(-\hbar\omega/\tau)]^3. \quad (S1)$$

(a) When $\tau \gg \hbar\omega$ the exponential in the square bracket may be expanded, leading to $[\dots] \cong \hbar\omega/\tau$ and to

$$\lambda_S \cong \exp(-\varepsilon_0/\tau) (\hbar\omega/\tau)^3. \quad (S2)$$

The equilibrium condition $\lambda_g = \lambda_S$ is, with λ_g from (30):

$$\frac{p}{\tau} \left(\frac{2\pi\hbar^2}{M\tau} \right)^{3/2} = \exp(-\varepsilon_0/\tau) (\hbar\omega/\tau)^3.$$

Cancelling $\hbar^3$ on both sides and multiplying both sides by $M^{3/2}\tau^{5/2}(2\pi)^{-3/2}$ yields (77).

(b) From (77), we form

$$\log p = \varepsilon_0/\tau - \frac{1}{2}\log \tau + \text{constant} .$$

If this is inserted into (20), we obtain

$$\frac{d}{dz} \log p = + \varepsilon_0/\tau^2 - \frac{1}{2}\tau = (\varepsilon_0 - \tau/2)/\tau^2 = L/\tau^2 ,$$

which implies $L = \varepsilon_0 - \tau/2$. This result may be understood by recalling from (23) that the latent heat is an enthalpy difference. We saw in (4.8) that in the high-temperature limit the energy of a single (one-dimensional) oscillator is $\langle \varepsilon \rangle \cong \tau$. A three-dimensional oscillator is equivalent to three one-dimensional oscillators, $\langle \varepsilon \rangle \cong 3\tau$. If the energy of the ground state is at $-\varepsilon_0$ instead of at zero: $\langle \varepsilon \rangle \cong -\varepsilon_0 + 3\tau$. For a solid, we may neglect the pv_s term in the enthalpy, hence the enthalpy per atom equals the energy, $H_s \cong -\varepsilon_0 + 3\tau$. For an ideal gas, $pv_g = \tau$, $U = 3\tau/2$, the enthalpy per atom is $H_g = U + pv_g = 5\tau/2$. From (23), if we replace H_l by H_s :

$$L = H_g - H_s = \varepsilon_0 - \tau/2 . \quad (S3)$$

Comment. On closer inspection, one finds that in (77) as well as in the latent heat expression (S3), ε_0 should be replaced by $\varepsilon_0 - 3\hbar\omega/2$. This is readily seen by carrying the high-temperature expansion of the exponential in (S1) one term further:

$$\begin{aligned} \lambda_s &\cong \exp(-\varepsilon_0/\tau) [(\hbar\omega/\tau) - \frac{1}{2}(\hbar\omega/\tau)^2]^3 \\ &= \exp(-\varepsilon_0/\tau) (\hbar\omega/\tau)^3 [1 - \frac{1}{2}(\hbar\omega/\tau)]^3 \\ &\cong \exp(-\varepsilon_0/\tau) [1 - (3/2)(\hbar\omega/\tau)] (\hbar\omega/\tau)^3 \end{aligned}$$

But $1 - 3\hbar\omega/2\tau \cong \exp(-3\hbar\omega/2\tau)$, hence

$$\lambda_s \cong \exp[-(\varepsilon_0 - 3\hbar\omega/2)/\tau] (\hbar\omega/\tau)^3 ,$$

which differs from (S2) by having ε_0 replaced by $\varepsilon_0 - 3\hbar\omega/2$, as claimed.

Similarly, in expanding (4.7), one should really write

$$\begin{aligned}\langle \varepsilon \rangle &= \frac{\hbar\omega}{(\hbar\omega/\tau) + \frac{1}{2}(\hbar\omega/\tau)^2} = \frac{\tau}{1 + \hbar\omega/2\tau} \\ &\cong \tau(1 - \hbar\omega/2\tau) = \tau - \frac{1}{2}\hbar\omega \quad .\end{aligned}$$

Note that $\frac{1}{2}\hbar\omega$ is just the zero-point energy of a harmonic oscillator, and that in the derivation of (4.7) we had chosen the zero of the energy scale at the ground state, rather than at the zero of the potential energy. If the zero of the energy is chosen at the zero of the potential energy the correction term $\frac{1}{2}\hbar\omega$ drops out. Similarly, (77) is correct as written, if $-\varepsilon_0$ is understood to be the energy of the state of zero potential energy of the oscillator in the solid, rather than the true ground state.

10.5) GAS-SOLID EQUILIBRIUM

(a) The N_s -term in (78) is evidently the free energy (for $\sigma = 0$) of N_s atoms with energy $-\varepsilon_0$ each, while the N_g -term is the free energy of an ideal gas of N_g atoms in a volume V -- see (6.24).

(b) Insert $N_s = N - N_g$ into (78). The minimum free energy then occurs when

$$\begin{aligned}(\partial F / \partial N_s)_{\tau, V} &= \varepsilon_0 + \tau [\log(N_g / V n_Q) - 1] + \tau \\ &= \varepsilon_0 + \tau \log(N_g / V n_Q) = 0 \quad .\end{aligned}$$

Solving for N_g yields (79).

(c) Insert (79) into the ideal gas law

$$p = N_g \tau / V = n_Q \tau \exp(-\varepsilon_0 / \tau) \quad .$$

10.6) THERMODYNAMICS OF THE SUPERCONDUCTING TRANSITION

(a) We saw in (8.56a) that the free energy difference per unit volume between the normal and the superconducting phase below $\tau = \tau_c$ is given by

$$[F_N(\tau) - F_S(\tau)]/V = B_C^2(\tau)/2\mu_0 \quad . \quad (8.56a)$$

The result (80) for the entropy difference follows by inserting this into (3.49), $\sigma = -(\partial F/\partial \tau)_{N,V,B}$. The vanishing of the entropies for $\tau \rightarrow 0$ implies

$$\left. \frac{dB_C}{d\tau} \right|_{\tau=0} = 0 \quad , \quad (S1)$$

that is, the B_C -versus- τ curve must have a horizontal tangent in the limit $\tau \rightarrow 0$. See Fig. 8.17.

(b) At $\tau = \tau_c$ superconducting and normal phases are in equilibrium, hence their free energies must be equal:

$$F_S(\tau_c) = F_N(\tau_c) \quad . \quad (S2)$$

(1) Because of (3.49), $\sigma_S(\tau_c) = \sigma_N(\tau_c)$ implies that also $(\partial F_S/\partial \tau) = (\partial F_N/\partial \tau)$ at $\tau = \tau_c$. Together with (S2) this implies that the two free energy curves do not cross but merge. (2) Because $U = F + \sigma\tau$, equality of both F and σ for the two phases at $\tau = \tau_c$ implies equality of U_S and U_N as well. (3) For any phase transition, the latent heat L is given by $L = \tau\Delta\sigma$, in extension of (13). $\Delta\sigma = 0$ implies $L = 0$. When the transition is carried out in a magnetic field, $\Delta\sigma$ is finite, and the latent heat per unit volume is obtained by multiplying (80) by $-\tau$:

$$L = (\sigma_N - \sigma_S)\tau/V = - \frac{B_C\tau}{\mu_0} \frac{dB_C}{d\tau} \quad .$$

We have chosen signs so as to make L positive.

(c) Eq. (81) is obtained by inserting (80) into the general relation (3.17), $VC_V = \tau(\partial\sigma/\partial\tau)$, between heat capacities and entropies. We have multiplied by V to make C the heat capacity per unit volume.

For $\tau \ll \tau_c$, the heat capacity of the superconductor becomes negligible besides that of the normal phase:

$$\Delta C \rightarrow -C_N \rightarrow -\gamma\tau \quad .$$

If this is inserted into (81) and a factor $-\tau$ is cancelled on both sides, the result may be written

$$\begin{aligned} \gamma &= -\frac{1}{2\mu_0} \left. \frac{d^2(B_C^2)}{d\tau^2} \right|_{\tau=0} = -\frac{1}{\mu_0} \left. \frac{d}{d\tau} B_C \frac{dB_C}{d\tau} \right|_{\tau=0} \\ &= -\frac{1}{\mu_0} \left[\left(\frac{dB_C}{d\tau} \right)^2 + B_C \frac{d^2 B_C}{d\tau^2} \right]_{\tau=0} . \end{aligned}$$

Because of (S1), this reduces to (82).

10.7) SIMPLIFIED MODEL OF THE SUPERCONDUCTING TRANSITION

We abbreviate $u = \tau/\tau_c$, in which case, from (83), $B_C^2 = B_{C0}^2(1-2u^2+u^4)$. Then (80) and (81) may be written

$$\begin{aligned} (\sigma_S - \sigma_N)/V &= \frac{1}{2\mu_0\tau_c} \frac{d(B_C^2)}{du} \\ &= \frac{2B_{C0}^2}{\mu_0\tau_c} (u^3 - u) = \frac{2B_{C0}^2}{\mu_0\tau_c} \left[\left(\frac{\tau}{\tau_c} \right)^3 - \left(\frac{\tau}{\tau_c} \right) \right] ; \quad (S1) \end{aligned}$$

$$C_S - C_N = \frac{\tau}{V} \frac{\partial}{\partial \tau} (\sigma_S - \sigma_N) = \frac{u}{2\mu_0\tau_c} \frac{d^2(B_C^2)}{du^2} .$$

If we insert $d^2(B_C^2)/du^2 = B_{C0}^2(12u^2-4)$:

$$C_S - C_N = \frac{2B_{c0}^2}{\mu_0 \tau_c} \left[3 \left(\frac{\tau}{\tau_c} \right)^3 - \left(\frac{\tau}{\tau_c} \right) \right] . \quad (S2)$$

If C_N is linear in τ , and C_S vanishes faster than linearly as $\tau \rightarrow 0$, then the τ^3 -terms in (S1) and (S2) clearly belong to the superconducting phase while the τ^1 -terms belong to the normal phase. If we define

$$\gamma = 2B_{c0}^2 / \mu_0 \tau_c^2 ,$$

as in Problem 10.6, we obtain the four relations

$$\begin{aligned} \sigma_S / V &= \gamma \tau^3 / \tau_c^2 , & \sigma_N / V &= \gamma \tau , \\ C_S &= 3\gamma \tau^3 / \tau_c^2 , & C_N &= \gamma \tau . \end{aligned}$$

Note that $C_S(\tau_c) = 3C_N(\tau_c)$.

The latent heat per unit volume L follows from

$$L = (\sigma_N - \sigma_S) \tau / V = \gamma \tau^2 [1 - (\tau / \tau_c)^2] .$$

It vanishes both at $\tau = 0$ and $\tau = \tau_c$.

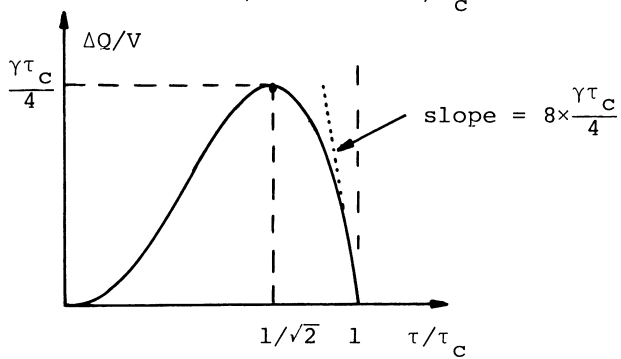
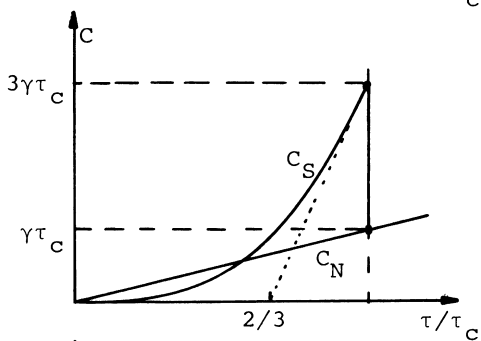
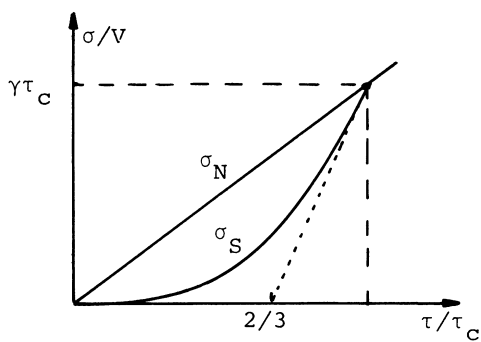
For plots of the σ 's, the C 's and of L see the next page.

10.8) FIRST-ORDER CRYSTAL TRANSFORMATION

(a) The energy of the phonons in the low temperature limit $T \ll \theta$ was given in (4.46), which may be written $U = c\tau^4$. If we understand the letter U to designate the energy density rather than the energy itself, then the proportionality constant c , with the help of (4.44), may be written

$$c = 3\pi^4 N / 5 (k_B \theta)^3 V = \pi^2 / 10 (\hbar v)^3 . \quad (S1)$$

The Helmholtz free energy density F of the phonons is most easily obtained through the relation (3.52),



$$F = U - \tau \sigma = U + \tau (\partial F / \partial \tau)_V .$$

With $U = c\tau^4$ one finds readily by inspection that the phonon contribution of the free energy density is

$$F = - c\tau^4/3 = - \pi^2 \tau^4 / 30 (\hbar v)^3 ,$$

leading to a total free energy density

$$F(\tau) = U(0) - c\tau^4/3 . \quad (S2)$$

The values of $U(0)$ and of c differ for the two crystal structures.

(b) At the transformation temperature τ_c we must have $G_\alpha(\tau_c) = G_\beta(\tau_c)$. If we neglect the very small effects of any volume changes, this condition becomes $F_\alpha(\tau_c) = F_\beta(\tau_c)$. Inserting (S2) and solving for τ_c^4 yields

$$\tau_c^4 = 3[U_\beta(0) - U_\alpha(0)]/(c_\beta - c_\alpha) . \quad (S3)$$

Inserting the values (S1) for the c 's yields (84).

(c) As in the liquid-gas transformation, the latent heat is, strictly speaking, an enthalpy difference -- see (23). If we again neglect the very small effects of any volume changes, the latent heat becomes simply the energy difference between the two phases at τ_c :

$$\begin{aligned} L &= H_\beta(\tau_c) - H_\alpha(\tau_c) \cong U_\beta(\tau_c) - U_\alpha(\tau_c) \\ &= U_\beta(0) - U_\alpha(0) - (c_\beta - c_\alpha)\tau_c^4 = 4[U_\beta(0) - U_\alpha(0)] , \end{aligned}$$

where in the last line we used (S3).